

Extraction validations for anthro RO SR

- Check and fix “overlapping data points” question
 - o If 0: within record, do data points overlap in outcome AND exposure?
 - o Look at instance descriptions
 - o Maybe fix manually

Columns: effect_multiple, effect_multiple_desc, and instance_description

-Description, instance_description, can be inspected to detect and duplicity (eg., adjusted vs adjusted). Now, the only issue with that is there is no standardization with descriptors, so that will be tough and manual.

-So, we can either compare all rows for duplicity, except extracted OR/RR values. Or we check the data in the 2x2 table for duplicity.

-After further consideration and some testing, it will be difficult to programmatically fix this one. Manual work recommended.

- Other within-instance/field-to-field validations
 - o Compare crude/calculated effect size to adjusted/extracted effect size?

Columns: effect_estimate, effect_lower, and effect_upper

-What is the goal of this validation?

-We want to see if there's a significant difference between adjusted vs unadjusted values.

-1. Check for adjusted or unadjusted 2. Same outcome, everything else should be the same. 3. No more than multiple of three or divisible by three. 4. The effect size should be inside the confidence interval, not outside.

- o Compare outcome names/descriptions with recorded acause/cause_id?

Columns: outcome, out_name, out_gbd_cause

-We need the full list to cross-compare. Data can potentially be extracted from the metadata df, check with Vinnie.

- Check the range/size of exposure groups:

- o extreme/unbelievable values

Columns: exp_lower_1, ref_lower_1, exp_upper_1, ref_upper_1

-From an initial inspection only the 1st exposure cohorts have data.

-Special characters (<, >, >=, =<) Have to be dealt with. Check with vinnie if these need any special attention. I'm curious why these special characters were encouraged to be extracted as well.

1. Post-processing to apply special characters. (e.g. <20, minus 0.01, >=23 we can rid of the symbol) 2. Get rid of the special symbols.

Extreme values: 10 and 60, checking for impossible values.

Lower should be lower than upper bound and vice versa. Do this separately for the exposure groups, reference and exposure.

- overlapping ranges among the alternative and reference groups, etc.

1. Compare exposure labels vs exposure measurements, making sure that these are correct.

2. If no measure start or end is given, compare the lower and upper. (CHECK AGAIN WITH PAN AND VINNIE ABOUT THIS ONE)

- List/compare all exposure groups by source to identify typos

Check with Vinnie

- Check the pattern of instance number questions?

Done after formatting the data. Ignore it for now.

- Check/flag certain fields for backfilling

- e.g., outcome-specific questions that we developed later

Example: GDM, whether preexisting subjects were included or not. Pass this for now. Modelers can do this in later post-processing.

- Unimportant fields?

For future projects for the moment? Once we finish this first round. Do it at the end.

Outputs

- List of errors fixed
- List of errors that need manual attention
 - Inform future extractions
- Forest plots – exposure x effect size grouped by outcome?
 - Other groupings?
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